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Network Anomaly Detection Algorithm Based on Deep Learning and Data Mining

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ABSTRACT

In this paper, we propose a new detection method for network anomaly detection, a key problem in the field of cyber security, which combines deep learning feature extraction and DBSCAN clustering algorithm. First, in terms of data processing and feature learning, we employ deep learning models CNN and RNN to automatically extract useful features from complex network traffic data. These advanced features can represent network behavior more accurately and provide richer information for subsequent clustering and anomaly detection. Then, using the DBSCAN algorithm, we can not only effectively identify normal behavioral patterns, but also accurately detect anomalies and potential threats. In the experimental part, by evaluating on the publicly available KDD Cup 99 dataset, the method in this paper demonstrates its superiority in key metrics such as accuracy and recall.

CCS CONCEPTS

• **Security and privacy** → Network security; • **Networks** → Network algorithms.

KEYWORDS

Network anomaly detection method, Deep learning feature extraction, DBSCAN clustering algorithm, data process, Identify behavioral pattern

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1 INTRODUCTION

With the rapid development of information technology and the popularization of Internet applications, network security has become an indispensable part of modern society. However, this dependence also brings serious security challenges [1-2]. Cyber threats, such as malware, phishing attacks, and DDoS attacks, not only threaten the data privacy and security of individual users, but also pose a

significant risk to the security of enterprises and the state. Therefore, effective cyber threat detection is crucial for maintaining a secure cyber environment. However, the emergence of new and sophisticated attacks makes traditional security measures often have limited effectiveness, and many existing security solutions rely on known threat signatures and cannot effectively detect zero-day attacks and unknown threats. In addition, with the explosive growth of network data, traditional methods are inefficient in handling massive amounts of data and cannot meet the demand for real-time detection [3-4].

In order to overcome these limitations, techniques of machine learning and data mining have been introduced in cyber threat detection. Machine learning provides the ability to learn from data and predict future behavior, while data mining can reveal hidden patterns and associations in data. These techniques automate the process of learning and recognizing complex attack behaviors from historical data, providing a more flexible and effective detection method. Deep learning, in particular, has achieved significant results in areas such as image processing and speech recognition, and has shown great potential for processing complex data and extracting deep features in cyber threat detection.

Given the continuous evolution of cyber threats and the limitations of existing methods, this research aims to develop a network anomaly detection algorithm based on deep learning and data mining. The goal of this research is to design and implement a system that can effectively identify and classify cyber threats, in particular, to automatically extract meaningful features from network data using deep learning, and to cluster analyze network behaviors using DBSCAN data mining techniques to effectively identify anomalous patterns and potential threats. In this way, we hope to improve the accuracy and speed of detection in order to effectively respond to complex and changing cyber threats.

The structure of this thesis is divided into four main sections. The first is the introduction, which provides the context for the entire study, discussing the severity of cyber threats and the limitations of traditional approaches, while emphasizing the potential of deep learning and data mining in overcoming these challenges. The next chapters will delve into the technical implementation, including data processing, feature learning, anomaly detection strategies, and performance evaluation, with the aim of experimentally validating the effectiveness of the proposed approach and analyzing it in comparison with existing methods. Finally, this thesis will summarize the research results and discuss possible directions for improvement and future research efforts.

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2 DATA PROCESSING AND DEEP LEARNING FEATURE LEARNING

In the research and practice of cyber threat detection, data processing and feature learning are the cornerstones for realizing an efficient detection system. This chapter is dedicated to introducing and exploring the entire process from data collection and preprocessing to deep learning model application. This includes detailing how data is collected and prepared, appropriate deep learning models are selected and trained, and how powerful features are extracted from these models for subsequent threat identification and classification. By carefully analyzing these key steps, we aim to build a robust detection framework that can effectively identify and respond to diverse cyber threats.

2.1 Data collection and pre-processing

High-quality and relevant data is a prerequisite for constructing effective models, while appropriate preprocessing methods can significantly improve data availability and model performance. This section will discuss in detail the key preprocessing steps such as data collection, cleaning, normalization, and feature extraction.

Data collection is the initial phase of network threat detection system design, which involves collecting data from multiple sources such as network traffic, system logs, application logs, and so on. This data usually contains various types of network metrics such as IP addresses, port numbers, protocol types, packet sizes, timestamps, and so on. The collection phase is immediately followed by data cleansing, which aims at removing redundant, incomplete, or irrelevant records. For example, it may be necessary to remove syslog information that is not relevant to network traffic, or to cull duplicate network flow records. For missing values, depending on their importance and distribution in the dataset, there are options to remove the corresponding records, populate the mean/median, or process them using more sophisticated interpolation methods.

Data standardization is another key step aimed at converting data of different magnitudes to a uniform standard and avoiding over-biasing the model towards features with larger values. The most commonly used standardization techniques include min-max normalization and Z-score normalization. Min-max normalization scales all features to between 0 and 1 by the following equation:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

Z-score standardization then transforms the data into a distribution with a mean of 0 and a standard deviation of 1, which is given by the formula:

$$Z = \frac{X - \mu}{\sigma} \quad (2)$$

Here, X is the original data, X_{min} , X_{max} represent the minimum and maximum values of the data, respectively, μ is the mean value of the data, and σ is the standard deviation. These standardization methods help to ensure the reasonable allocation of the weights of each feature in the training process of the model, thus improving the model's generalization ability to new data.

Feature extraction is the process of converting raw data into a form that can be effectively processed by a model. In the context of cyber threat detection, feature extraction is especially critical as it directly affects the performance of the detection algorithm. Deep

learning offers powerful feature extraction capabilities, with Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) being the most common choices.

When processing network traffic data, CNNs are able to capture spatial patterns, such as combinations of features in a sequence of packets. A basic CNN structure consists of a convolutional layer, an activation layer, and a pooling layer. The main role of the convolutional layer is to extract the local features of the input data through the convolutional kernel. Assuming that the input data is X and the convolution kernel is K , the convolution operation can be expressed as:

$$F(i, j) = (X * K)(i, j) = \sum_m \sum_n X(m, n) K(i - m, j - n) \quad (3)$$

Where $F(i, j)$ is the element in the feature map obtained after the convolution operation. Next, activation layers such as ReLU introduce nonlinearity into the network to enhance the model representation. The pooling layer is then used to reduce the feature dimensions and strengthen the robustness of the features.

The RNN is particularly effective for sequential data, such as time-series network traffic. Its ability to capture temporal dynamics in the data is useful for modeling behavioral patterns in network traffic. The key to an RNN is its hidden state, which captures information about the sequence up to the present. The basic RNN unit can be represented as:

$$h_t = \sigma(W_{xh}x_t + W_{hh}h_{t-1} + b_h) \quad (4)$$

Where h_t is the hidden state at time step t , x_t is the input at time step t , W_{xh} and W_{hh} are the weight matrices, b_h is the bias term, and σ is the activation function.

Feature extraction by these deep learning models results in highly representational and high abstraction level features that capture not only the basic statistical properties of the data, but also complex patterns and structural information.

3 DBSCAN CLUSTERING AND ANOMALY DETECTION

This section will focus on how to apply the DBSCAN clustering algorithm for network anomaly detection, in particular how to utilize the deep learning features extracted in the previous chapter to enhance the accuracy and efficiency of the detection. DBSCAN, as a density-based clustering method, is particularly suited for dealing with complex and high-dimensional data, making it a powerful tool for detecting anomalous behaviors in networks.

3.1 Principle of the DBSCAN algorithm

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a well-known clustering algorithm that classifies clusters or groups based on the 'density' in a data set. Unlike distance-based clustering methods, DBSCAN does not require the number of clusters to be specified in advance, but rather determines the formation of clusters based on the structure and distribution of the data itself. This makes DBSCAN particularly effective when dealing with data with complex structure or noise. The core idea of DBSCAN is to identify high-density regions separated by low-density regions that form the bulk of the clusters, while points in the low-density regions are considered noise or anomalies.

The DBSCAN algorithm operates based on two main parameters: neighborhood radius (ϵ) and minimum number of points (MinPts). Neighborhood radius (ϵ) defines the size of the region surrounding a point, while the minimum number of points (MinPts) is the minimum number of points that need to be present in a point's ϵ -neighborhood in order for the point to be a core point. Specifically, a point is labeled as a core point if it has at least MinPts of points (including the point itself) within its ϵ -neighborhood. Core points are the basic elements that form clusters. Boundary points are those points that are not core points but are within the ϵ -neighborhood of a core point, while all points that are not core points nor boundary points are considered as noise. In this way, DBSCAN classifies data points into three categories: core points, boundary points and noise points.

After defining the core and boundary points, DBSCAN starts forming clusters. Starting from any core point, the algorithm adds all the direct density reachable points of that point to the same cluster. This process is recursive, i.e., for each new point added to the cluster, the algorithm also adds its density-accessible points to the same cluster. This process continues until no new points can be added to the cluster. It is worth noting that since boundary points may be in the ϵ -neighborhood of multiple core points, they may be assigned to multiple clusters by the algorithm, but in the final clustering result, each boundary point can only belong to one cluster. In addition, the noise points do not belong to any cluster; they are considered as separate, irregular points.

The advantage of the DBSCAN algorithm is that it makes no assumptions about the shape of the clusters, is able to recognize clusters of arbitrary shapes, and is able to deal with noise and anomalies efficiently. This property makes it well suited for applications such as cyber threat detection, where anomalous behavior often manifests itself in patterns significantly different from normal behavior. However, the performance of DBSCAN is highly dependent on the choice of ϵ and MinPts. Inappropriate parameters may lead to excessive noise or excessive cluster merging. Therefore, careful parameter tuning and selection based on domain knowledge are crucial in practical applications. By incorporating deep learning feature learning approaches, as described in the previous chapter, DBSCAN is able to perform in higher dimensional and more complex feature spaces, further improving the accuracy and efficiency of cyber threat detection.

3.2 Anomaly Detection Strategy

In the context of cyber threat detection, the goal of an anomaly detection strategy is to accurately identify activities that do not conform to normal behavioral patterns that may point to potential threats or attacks. Using the properties of the DBSCAN clustering algorithm, we can design an effective anomaly detection strategy that is based on the assumption that clusters formed from normal behaviors are separated from anomalous behaviors. With the advanced features extracted by deep learning in Section II, we will be able to identify and classify behavioral patterns in the network at a finer granularity.

First, the clusters generated by DBSCAN during the clustering process are assumed to represent normal behavioral patterns in network traffic. These clusters are formed by identifying dense regions

in a multidimensional feature space in which the points of each cluster are closely connected, reflecting some degree of behavioral consistency. On this basis, points that are not categorized into any cluster—usually isolated or located in low-density regions—are considered anomalies. These points may represent anomalous behavior in the network, such as an attack or system failure. In DBSCAN, these points are known as noise points. Considering these noise points as potential threats, we can analyze and investigate them further.

However, simply considering all noise points as anomalies may lead to high false alarm rates. To improve the accuracy of detection, the characteristics and context of these noise points need to be further analyzed. This includes examining their location in the feature space, their relationship with nearest neighbor clusters, and their behavior over time. For example, if a noise point is close to a large normal-behavior cluster in feature space, it may be only a slight deviation rather than a true anomaly. Conversely, if a noise point is far from any cluster, then it is more likely to be a true anomaly.

In addition, given the dynamic nature of the network environment, our anomaly detection strategy also needs to be adaptive. This means that the clustering model should be updated periodically to reflect changes in network behavior patterns. In the detection strategy, we also utilize supervised learning methods to use known instances of normal and abnormal behaviors for training, which helps determine whether unlabeled noise points are normal or abnormal. In this way, the detection strategy can be optimized over time to adapt to changes in the network environment. The algorithmic flow of the anomaly detection strategy is shown in Figure 1.

4 EXPERIMENTAL RESULTS

In this section, we detail the experimental setup for implementing the cyber threat detection system, the evaluation metrics used, and the performance results on specific datasets. These experiments aim to validate the effectiveness of the anomaly detection strategy based on DBSCAN clustering and deep learning features proposed in the previous sections.

4.1 Experimental setup

Experimental design is an important part of evaluating the performance of a detection system. First, a dataset containing a wide range of network behaviors (both normal behaviors and various types of threats) needs to be selected or constructed. For this purpose, we chose a widely used public dataset: the KDD Cup 99 dataset. This dataset contains network connection records for both normal behavior and various types of attacks, and is one of the commonly used benchmark datasets in cybersecurity research. The KDD Cup 99 dataset consists of millions of network connection records, each of which contains detailed information about a single connection, such as duration, protocol type, service type, etc., as well as a set of statistical characteristics derived from network traffic analysis. These records are labeled as normal or a specific type of attack such as DoS, U2R, R2L, and Probe. Next, we divide the dataset into a training set and a test set. The training set is used to build and optimize the detection model, including the learning

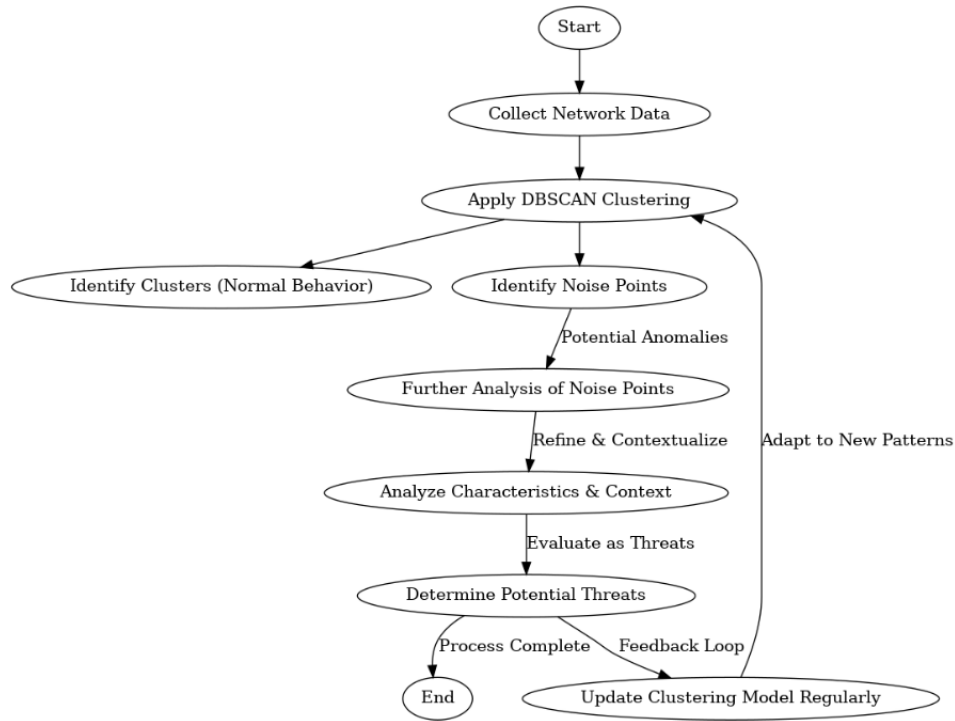


Figure 1: Schematic diagram of anomaly detection strategy implementation

of deep learning features and the training of the DBSCAN clustering model. The test set is then used to evaluate the detection performance of the model.

4.2 Assessment of indicators

In order to comprehensively evaluate the performance of the network threat detection system, we use a variety of evaluation metrics, mainly including accuracy, recall and F1 score. These metrics reflect the performance of the model from different perspectives, respectively:

- (1) Precision: refers to the proportion of true anomalies among all instances recognized as anomalies by the model. It measures the accuracy of the model in recognizing threats.
- (2) Recall: refers to the proportion of all instances of real anomalies that are correctly recognized by the model. It measures the ability of the model to capture threats.

For the comparison experiments, the following network threat detection algorithms were selected: network anomaly detection methods based on machine learning algorithms [5] and network anomaly detection methods based on self-encoders [6]. All algorithms are evaluated on the same dataset, which ensures the fairness of the evaluation. In our experiments, we use the same data preprocessing steps to ensure that all algorithms have the same quality of input data. For each method, we adjusted its parameters to achieve its optimal performance.

4.3 Analysis of experimental data

In this experiment, we compare three different approaches for network anomaly detection: a traditional machine learning based approach (referred to as literature [5]), a self-encoder based approach (referred to as literature [6]), and the algorithm proposed in this paper. The main objective of the experiment is to evaluate and compare the performance of these methods in terms of accuracy (Precision). As can be seen from the experimental results in Figure 2, the Precision of all three methods fluctuates with the change of conditions or thresholds. Overall, the accuracy of all the methods shows a certain degree of volatility, which may be due to the complexity and variability of cyber threats. Among the fluctuations, we can see that the algorithm proposed in this paper slightly outperforms the other two methods in most cases, showing slightly higher accuracy. Literature [5] and literature [6], while performing well in some conditions, show some fluctuations in the overall trend, suggesting that they may be more sensitive to specific types of network behaviors or threats. In contrast, the method proposed in this paper not only maintains a relatively stable accuracy under most conditions, but also shows a slight advantage across the entire range of experiments.

Recall is an important index for evaluating the performance of the model, which measures the ability of the model to correctly identify real anomalies and is crucial for evaluating whether the system can effectively avoid underreporting. Based on the experimental data in Figure 3, we can see that the three different network anomaly detection methods all show some volatility in the recall rate with changes in the environment or the set threshold, which

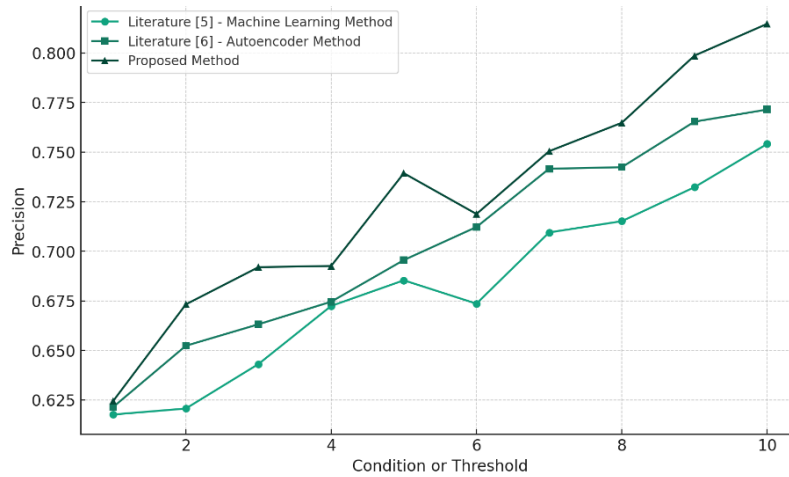


Figure 2: Accuracy of different algorithms

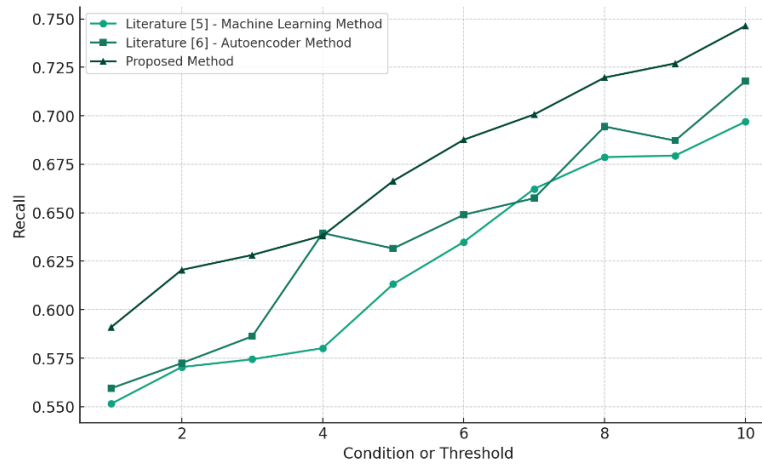


Figure 3: Recall of different algorithms

reveals the adaptability of each algorithm to different network environments and its sensitivity to anomaly detection.

Specifically, the methods introduced in literature [5] and literature [6] are able to achieve high recall rates under specific conditions, but overall, the recall rates of these two methods fluctuate more, indicating that they may show better performance in identifying some specific types of network behaviors or threats, however, they may suffer from a lack of stability in other cases. In contrast, the algorithm proposed in this paper exhibits a more stable recall rate overall, especially under variable conditions still able to maintain a relatively smooth performance, and presents a superior performance to the other two methods in most cases. This suggests that although all methods have challenges in facing complex and changing network environments, the network anomaly detection method proposed in this paper has certain advantages in maintaining the stability of the recall rate and avoiding underreporting, which reflects its potential and effectiveness in identifying network threats in practical applications.

5 CONCLUSION

In this paper, we propose a new detection method for network anomaly detection, a key problem in the field of cyber security, which combines deep learning feature extraction and DBSCAN clustering algorithm. By comparing and analyzing the performance with other popular methods, this study demonstrates the effectiveness and potential of the proposed method in network threat detection. First, for data processing and feature learning, we employ deep learning models, such as CNN and RNN, to automatically extract useful features from complex network traffic data. These advanced features can represent network behavior more accurately and provide richer information for subsequent clustering and anomaly detection. Then, using the DBSCAN algorithm, we can not only effectively identify normal behavioral patterns, but also accurately detect anomalies and potential threats.

In the experimental part, by evaluating on the publicly available KDD Cup 99 dataset, the method in this paper demonstrates its superior performance in key metrics such as accuracy and recall.

Despite the superior performance shown by this paper's method, false positives and omissions are still challenges that require further research and improvement. In addition, real-time performance, stability and adaptability are also important factors to be considered for practical deployment. Future work will focus on further improving the detection performance, especially on enhancing feature learning and anomaly detection through more advanced deep learning models and clustering algorithms. We also plan to validate the effectiveness and robustness of the approach on larger and more complex real-world datasets, as well as exploring how to combine the approach in this paper with other security mechanisms to form a more comprehensive cybersecurity solution.

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